

MARCUS OSEI

Email: marcus.osei@gmail.com · GitHub: github.com/marcosei · Portfolio: marcosei.dev

DESIRED ROLE: Backend Software Engineer

EDUCATION

Self-directed learning (no formal CS degree)

Completed: freeCodeCamp Back End Development and APIs Certification (2022)

Completed: The Odin Project Full Stack JavaScript Path (2021–2022)

Completed: CS50x: Introduction to Computer Science, Harvard / edX (2021)

WORK EXPERIENCE

Freelance Web Developer (Jan 2022 – Present)

- Build and maintain backend systems for three small business clients
 - Technologies used: Node.js, Express, PostgreSQL, REST APIs, Heroku
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PROJECTS

LinkLocal — Community Events Platform (2023)

Live at: linklocal.app

- Full-stack web application connecting local communities with neighbourhood events
- Backend: Node.js / Express REST API with JWT authentication and role-based access control
- Database: PostgreSQL with normalised schema; handled concurrent reads with connection pooling
- Deployed on Heroku with automated CI/CD via GitHub Actions
- 340 registered users as of March 2024; handles approximately 1,200 API requests per day
- Implemented rate limiting, input validation, and error handling middleware

BudgetBee — Personal Finance Tracker (2022)

Archived at: github.com/marcosei/budgetbee

- Node.js / Express backend with SQLite database
- User authentication with bcrypt password hashing and session tokens
- RESTful API consumed by a vanilla JavaScript frontend
- Wrote unit tests using Jest; achieved 74% code coverage
- Deployed on a VPS (DigitalOcean Droplet) with Nginx reverse proxy and SSL

StockAlert — Price Notification Service (2022)

github.com/marcosei/stockalert

- Node.js cron job service polling a public stock API every 15 minutes
 - Sends email notifications via SendGrid when user-defined thresholds are crossed
 - Persistent user configuration stored in PostgreSQL
 - Designed as a stateless service — horizontally scalable
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TECHNICAL SKILLS

Languages: JavaScript (Node.js), SQL, Python (scripting only), Bash

Frameworks: Express.js

Databases: PostgreSQL, SQLite

Infrastructure: Heroku, DigitalOcean, Nginx, GitHub Actions

Testing: Jest

Other: REST API design, JWT, bcrypt, rate limiting, cron jobs, SendGrid